WiSAR FIELD OPERATIONAL CHECKLIST

Sweep Width (W), Coverage (C), POD Calculation & Bayesian POA Updating

OPERATIONAL COMMAND MANDATE: Search efficiency is maximized when ground teams are deployed according to Probability Density (Pden = POA / Area) and Probability of Success Rate (PSR = W × V × Pden). Do not deploy resources uniformly; calculate Coverage (C) and update segment Probability of Area (POA) after every search assignment using Bayesian rules.

1. MATHEMATICAL FORMULAS & FIELD CALCULATIONS

Metric	Formula / Equation	Field Definition & Key Parameters
Effective Sweep Width (W)	$W = \int f(x)\,dx$	Lateral distance where missed targets inside W equal detected targets outside W. Measured in meters/yards.
Total Track Line (TTL)	TTL = N_searchers × Speed (V) × Time (T)	Total distance swept by all searchers or drones in a task segment (e.g., 4 searchers × 2 km/h × 3 hrs = 24 km).
Swept Area (Z)	Z = TTL × W	Total effective area swept by search resources (e.g., 24 km × 0.05 km sweep width = 1.2 sq km).
Coverage Ratio (C)	C = Z / A = (TTL × W) / A	Thoroughness of search. For parallel grid sweeps with searcher spacing S, C simplifies to C = W / S.
Probability of Detection (POD)	POD = 1 - e^(-C)	Random search model detection probability. At C = 0.7, POD ≈ 50%. At C = 1.0, POD ≈ 63%. At C = 2.0, POD ≈ 86%.
Probability of Success (POS)	POS = POA × POD	The actual probability of finding the subject in a segment during a specific task assignment.
Bayesian POA Shift	POA_after = POA_before × (1 - POD)	Updated segment POA after an unsuccessful search pass. Shifted probability is redistributed to remaining segments.

2. FIELD DETERMINATION OF EFFECTIVE SWEEP WIDTH (W)

Effective Sweep Width (**W**) varies dramatically based on vegetation density, subject posture/clothing, lighting, and fatigue. Search managers should determine **W** using one of three field calibration methods:

Method	Field Procedure	Operational Guidance
Visual Detection Profile (VDP)	Place a target profile (clothespins/objects) in representative terrain. Walk past at set distances and mark detection/miss.	Fastest field method for team leaders prior to deployment.
Average Detection Distance (ADD)	Walk toward a hidden target from 8 cardinal directions; measure detection distance for each. W ≈ 2 × ADD.	Provides a reliable mathematical approximation in open to medium vegetation.
Historical ISRID Lookup Tables	Reference validated ISRID sweep width benchmark tables for specific terrain, subject type, and searcher type.	Standard baseline when field calibration time is severely constrained.

3. SEARCH MANAGER STEP-BY-STEP FIELD WORKFLOW

[ ] Step 1: Segment & Calculate Pden	Divide search map into operational segments. Calculate Area (A) for each segment. Assign initial Probability of Area (POA) based on ISRID distance rings/behavior models. Calculate Probability Density: Pden = POA / Area.

[ ] Step 2: Prioritize Tasks by PSR	Rank segments by Probability of Success Rate: PSR = W × V × Pden. Assign fastest, high-capability assets (dogs, hasty teams) to segments with highest PSR.
[ ] Step 3: Set Team Spacing (S)	For parallel grid sweeps, set searcher spacing S based on target Coverage C. For C = 0.7 (50% POD tipping point), set spacing S = W / 0.7 ≈ 1.4 × W.
[ ] Step 4: Execute & Log Track Length	Log actual Total Track Line (TTL) using GPS tracks. Calculate total swept area Z = TTL × W.
[ ] Step 5: Calculate Actual POD	Calculate segment Coverage C = Z / A. Determine POD using POD = 1 - e^(-C) (or calibrated visual POD lookup curves).
[ ] Step 6: Execute Bayesian Update	If subject is not found, update segment POA: POA_after = POA_before × (1 - POD). Renormalize remaining segment POAs across the search area.

4. COVERAGE (C) VS. POD BENCHMARK LOOKUP TABLE

Coverage (C)	Theoretical POD (1 - e^-C)	Field Search Description	Resource Efficiency Rating
0.10	**9.5%**	Very low thoroughness / rapid hasty sweep along main trail	High Speed / Low POD
0.30	**25.9%**	Light search sweep; widely spaced team	Moderate Speed
0.50	**39.3%**	Medium search sweep	Good balance for large areas
0.70	**50.3%**	TIPPING POINT: Optimal balance of speed & POD	**MAX EFFICIENCY TIPPING POINT**
1.00	**63.2%**	Standard 100% coverage sweep (Spacing S = W)	Standard Grid Search
1.50	**77.7%**	Thorough grid search pass	Diminishing Returns Set In
2.00	**86.5%**	Very thorough double-grid search pass	High Effort / Low Efficiency
3.00	**95.0%**	Near-exhaustive search sweep	Exhaustive / Restrictive

5. SUBJECT CATEGORY SPATIAL BEHAVIOR QUICK-REFERENCE

Subject Profile	Key Behavioral Pattern	Highest Probability Search Zones	Recommended Search Asset
Dementia / Alzheimer's	Walks until stuck; 66% follow roads; highly directional heading.	Road edges, creeks/drainages (28%), briars/brush (33%). Within 15-30m of linear features.	Hasty vehicles, tracking dogs, thermal UAVs along roads/creeks.
Children (1-3 yrs)	Non-goal oriented; seeks small enclosed spaces when tired/scared.	50% found within 300m of PLS. Under sheds, thickets, parked vehicles, structures.	Ground teams checking structures, tight line sweeps around PLS.
Children (4-6 yrs)	Tries to get home; follows path of least resistance (animal trails).	Drainages, gentle downhill paths, clear trails.	Hasty teams along trails/creeks.
Children (7-12 yrs)	May panic ("trail running"); seeks shortcuts; attempts view-enhancing.	Trails, ridges, structures (50% seek shelter in outbuildings).	Fast trail hasty sweeps, drone sweeps of ridgelines.
Despondent / Suicidal	Goal-oriented toward privacy; seeks scenic or secluded locations.	Scenic overlooks, deep woods, structure interiors, vehicles.	Thermal drones, hasty search of high-lookout features.

6. BAYESIAN SEGMENT UPDATE WORKBOOK

Segment ID	Area (A)	Prior POA	Swept Dist (TTL)	Sweep W	Coverage (C)	Segment POD	Unsearched Factor (1-POD)	New Unnorm. POA	Normalized New POA
Segment A	1.5 km²	40%	12 km	0.05 km	0.40	33.0%	0.67	26.8%	34.5%
Segment B	2.0 km²	35%	28 km	0.05 km	0.70	50.3%	0.50	17.5%	22.5%
Segment C	1.0 km²	25%	0 km	0.05 km	0.00	0.0%	1.00	25.0%	32.2%
TOTALS	**4.5 km²**	**100%**	**40 km**	—	—	—	—	**Sum = 69.3%**	**Normalized = 100%**

> **COMMAND SUMMARY:** Always update probabilities immediately after teams complete segment sweeps. Do not over-search low Pden segments to high POD. Reallocate search resources to adjacent unsearched segments once current segment POD reaches the 50% tipping point (Coverage C ≈ 0.7).